

Routeur 1.3

1 Renommer le routeur

```
> enable
> conf t
> hostname routeur1.3
```

2 Assigner IP à l'interface 0/0

```
> enable
(config)# conf t
(config)# interface G0/0
(config-if)# no shutdown
(config-if)# exit
(config)# interface G0/0.10
(config-subif)# encapsulation dot1Q 10
(config-subif)# ip address 192.168.1.1 255.255.255.224
ipv6 address 2001:db8:1::1/64
(config-subif)# ip helper-address 192.168.0.3
(config-subif)# no shutdown
(config-subif)# exit
(config)# interface G0/0.20
(config-subif)# encapsulation dot1Q 20
(config-subif)# ip address 192.168.1.2 255.255.255.192
ipv6 address 2001:db8:2::1/64
(config-subif)# ip helper-address 192.168.0.3
(config-subif)# no shutdown
(config-subif)# exit
(config)# interface G0/1
(config-if)# ip address 172.10.0.1 255.255.255.252
ipv6 address 2001:db9:10::1/64
(config-if)# no shutdown
```



3 OSPF ipv4

```
enable
conf t
router ospf 1
network 192.168.1.0 0.0.0.31 area 10
```

```
network 192.168.2.0 0.0.0.63 area 10
network 172.10.0.0 0.0.0.3 area 0
router-id 1.1.1.3
exit
exit
wr
```

4 OSPF ipv6

```
enable
conf t
ipv6 unicast-routing
ipv6 router ospf 2
router-id 1.1.1.3
exit
interface G0/0.10
ipv6 ospf 2 area 0
exit
interface G0/0.20
ipv6 ospf 2 area 0
exit
interface G0/1
ipv6 ospf 2 area 0
exit
exit
wr
```

Routeur 1.4

```
enable
conf t
interface g0/1
ip address 172.10.0.2 255.255.255.252
ipv6 address 2001:db9:10::2/64
ip helper-address 192.168.0.3
no shutdown
exit

interface g0/0
ip address 172.20.0.1 255.255.255.252
ipv6 address 2001:db9:20::1/64
ip helper-address 192.168.0.3
```

```
no shutdown
exit

interface g0/2
ip address 172.30.0.1 255.255.255.252
ipv6 address 2001:db9:30::1/64
ip helper-address 192.168.0.3
no shutdown
exit
exit
wr
```

1 OSPF ipv4

```
enable
conf t
router ospf 1
network 172.10.0.0 0.0.0.3 area 0
network 172.20.0.0 0.0.0.3 area 0
network 172.30.0.0 0.0.0.3 area 0
router-id 1.1.1.4
exit
exit
wr
```

2 OSPF ipv6

```
enable
conf t
ipv6 unicast-routing
ipv6 router ospf 2
router-id 1.1.1.4
exit
interface G0/1
ipv6 ospf 2 area 0
exit
interface G0/0
ipv6 ospf 2 area 0
exit
interface G0/2
ipv6 ospf 2 area 0
exit
```

```
exit
wr
```

Routeur 1.5

```
enable
conf t
interface G0/0
ip address 172.20.0.2 255.255.255.252
ipv6 address 2001:db9:20::2/64
ip helper-address 192.168.0.3
no shutdown
exit

interface G0/1
ip address 172.40.0.1 255.255.255.252
ipv6 address 2001:db9:40::1/64
ip helper-address 192.168.0.3
no shutdown
exit

interface G0/2
ip address 172.50.0.1 255.255.255.252
ipv6 address 2001:db9:50:1/64
ip helper-address 192.168.0.3
no shutdown
exit
exit
wr
```

1 OSPF ipv4

```
enable
conf t
router ospf 1
network 172.20.0.0 0.0.0.3 area 0
network 172.50.0.0 0.0.0.3 area 0
network 172.40.0.0 0.0.0.3 area 0
router-id 1.1.1.5
exit
```

```
exit
wr
```

2 OSPF ipv6

```
enable
conf t
ipv6 unicast-routing
ipv6 router ospf 2
router-id 1.1.1.5
exit
interface G0/0
ipv6 ospf 2 area 0
exit
interface G0/1
ipv6 ospf 2 area 0
exit
interface G0/2
ipv6 ospf 2 area 0
exit
exit
wr
```

3 BGP

Routeur 1.6

```
enable
conf t
interface G0/2
ip address 172.30.0.2 255.255.255.252
ipv6 address 2001:db9:30::2/64
ip helper-address 192.168.0.3
no shutdown
exit

interface G0/1
ip address 172.40.0.2 255.255.255.252
ipv6 address 2001:db9:40::2/64
ip helper-address 192.168.0.3
```

```
no shutdown
exit
exit
wr
```

3 OSPF ipv4

```
enable
conf t
router ospf 1
network 172.40.0.0 0.0.0.3 area 0
network 172.30.0.0 0.0.0.3 area 0
router-id 1.1.1.6
exit
exit
wr
```

2 OSPF ipv6

```
enable
conf t
ipv6 unicast-routing
ipv6 router ospf 2
router-id 1.1.1.6
exit
interface G0/0
ipv6 ospf 2 area 0
exit
interface G0/1
ipv6 ospf 2 area 0
exit
interface G0/2
ipv6 ospf 2 area 0
exit
exit
wr
```

4 BGP

Routeur 2.4

```
enable
conf t
interface g0/0
ip address 192.168.0.2 255.255.255.240
ipv6 address 2001:dba:1::2/64
ip helper-address 192.168.0.3
no shutdown
exit

interface G0/1
no shutdown
exit

interface G0/1.30
encapsulation dot1Q 30
ip address 192.168.3.1 255.255.255.224
ipv6 address 2001:db8:3::1/64
ip helper-address 192.168.0.3
no shutdown
exit

interface G0/1.40
encapsulation dot1Q 40
ip address 192.168.4.1 255.255.255.128
ipv6 address 2001:db8:4::1/64
ip helper-address 192.168.0.3
no shutdown
exit
exit
wr
```

3 OSPF ipv4

```
enable
conf t
router ospf 1
network 192.168.0.0 0.0.0.15 area 0
network 192.168.3.0 0.0.0.31 area 20
network 192.168.4.0 0.0.0.127 area 20
```

```
router-id 2.2.2.4
exit
exit
wr
```

2 OSPF ipv6

```
enable
conf t
ipv6 unicast-routing
ipv6 router ospf 2
router-id 2.2.2.4
exit
interface G0/0
ipv6 ospf 2 area 0
exit
interface G0/1.30
ipv6 ospf 2 area 20
exit
interface G0/1.40
ipv6 ospf 2 area 20
exit
exit
wr
```

4 BGP

Routeur 2.5

```
enable
conf t
interface G0/2
ip address 172.50.0.2 255.255.255.252
ipv6 address 2001:db9:50::2/64 ##faute de frappe à adresse ip
ip helper-address 192.168.0.3
no shutdown
exit

interface G0/1
ip address 192.168.0.1 255.255.255.240
```



```
ipv6 address 2001:db8:0::1/64
ip helper-address 192.168.0.3
no shutdown
exit

interface G0/0
ip address 20.0.0.1 255.255.255.252
ipv6 address 2001:dba:1::1/64
ip helper-address 192.168.0.3
no shutdown
exit
exit
wr
```

1 OSPF ipv4

```
enable
conf t
network 172.50.0.0 0.0.0.3 area 0
network 192.168.0.0 0.0.0.15 area 0
router-id 2.2.2.5
exit
exit
wr
```

2 OSPF ipv6

```
enable
conf t
ipv6 unicast-routing
ipv6 router ospf 2
router-id 2.2.2.5
exit
interface G0/1
ipv6 ospf 2 area 0
exit
interface G0/2
ipv6 ospf 2 area 0
exit
exit
wr
```

2 BGP

```
enable
conf t
router bgp 65001
neighbor 20.0.0.2 remote-as 650002
neighbor 20.0.0.2 activate
```

3 nat pat

```
enable
conf t
interface g0/1
ip nat inside
no shutdown
exit
interface G0/0
ip nat outside
no shutdown
exit
ip nat inside source static tcp 192.168.0.6 80 20.0.0.1 80 (http)
ip nat inside source static tcp 192.168.0.6 443 20.0.0.1 443 (https)
ip nat inside source static tcp 192.168.0.5 137 20.0.0.1 137 (samba)
ip nat inside source static tcp 192.168.0.5 139 20.0.0.1 139 (samba)
ip nat inside source static tcp 192.168.0.5 138 20.0.0.1 138 (samba)
ip nat inside source static tcp 192.168.0.5 145 20.0.0.1 145 (samba)
ip nat inside source static tcp 192.168.0.8 143 20.0.0.1 143 (IMAP)
ip nat inside source static tcp 192.168.0.7 587 20.0.0.1 587 (smtp)

ip nat inside source static tcp 192.168.0.0 80 20.0.0.1 80
wr
```

Routeur 2.6

```
enable
conf t

interface g0/0
ip address 20.0.0.2 255.255.255.252
ipv6 2001:dba:1::2/64
ip helper-address 192.168.0.3
no shutdown
```

```
exit

interface g0/1
no shutdown
exit

interface g0/1.50
encapsulation dot1Q 50
ip address 192.168.5.1 255.255.255.240
ipv6 address 2001:db8:5::1/64
ip helper-address 192.168.0.3
no shutdown
exit

interface g0/1.60
encapsulation dot1Q 60
ip address 192.168.6.1 255.255.255.224
ipv6 address 2001:db8:6::1/64
ip helper-address 192.168.0.3
no shutdown
exit
exit
wr
```

3 OSPF ipv4

```
enable
conf t
router ospf 1
network 192.168.5.0 0.0.0.15 area 30
network 192.168.6.0 0.0.0.31 area 30
router-id 2.2.2.4
exit
exit
wr
```

2 OSPF ipv6

```
enable
conf t
ipv6 unicast-routing
ipv6 router ospf 2
```

```
router-id 2.2.2.6
exit
interface G0/1.50
ipv6 ospf 2 area 30
exit
interface G0/1.60
ipv6 ospf 2 area 30
exit
exit
wr
```

4 BGP

```
enable
conf t
router bgp 65002
neighbor 20.0.0.1 remote-as 65001
neighbor 20.0.0.1 activate

network 192.168.5.1 255.255.255.240 //
network 192.168.6.1 255.255.255.224 //

network 20.0.0.0 mask 255.255.255.252 !!
```

3 nat pat